

UZAIR ASHFAQ

📍 Valence, France 📞 +33 7 45 46 52 98 ✉ uzairashfaq85@gmail.com
🌐 uzairashfaq85 📧 uzairashfaq85 🌐 Portfolio

About Me

Master's student in Embedded Systems Security at Grenoble INP-Esisar with hands-on experience in embedded firmware, RFID/RTLS software, connected devices, and hardware-software integration. Skilled in MCU programming (STM32, PIC, ESP32), embedded Linux, RTL design, verification, and PCB development. Currently working as an Embedded Systems Engineer Intern at Acceliot on UHF RFID-based RTLS and software service platforms. Actively seeking full-time opportunities in embedded systems, firmware development, or secure connected devices upon completion of my Master's degree in September 2026.

Education

International Master in Embedded Systems Security (IMESS) Sep 2025 – Present
Grenoble INP-Esisar, Université Grenoble Alpes, Valence, France

BS in Electrical (Electronics) Engineering Sep 2020 – Aug 2024
COMSATS University Islamabad, Pakistan

Experience

Embedded Systems Engineer Intern—Acceliot, Feb 2026 – Present
Valence, France

- Developing a standalone Software Service Platform (SSP) for UHF RFID-based RTLS and DSAI systems.
- Designing a vendor-agnostic data model and interfaces to ingest raw RFID reader data, including RSSI, phase, timestamps, and antenna IDs.
- Implementing phase calibration, unwrapping, and phase-difference feature extraction for AoA and positioning indicators.
- Building scalable Python pipelines using Pandas, NumPy, and SciPy for data cleaning, structuring, and feature engineering.
- Applying and validating machine learning models using scikit-learn for detection and estimation tasks with clear evaluation metrics.
- Contributing to a reusable Linux-targeted software package, documentation, and demonstrator for deployment and validation.

Embedded Systems Engineer – Lean Automation Sep 2024 – Aug 2025
Abbottabad, Pakistan

- Developed embedded C firmware for STM32 and PIC microcontrollers, integrating UART, I2C, timers, and interrupt-driven control loops for industrial automation systems.
- Improved communication reliability and system responsiveness through robust peripheral integration and firmware debugging.
- Designed and revised 4-layer PCBs in Altium Designer, improving signal integrity and reducing bench-level debugging during hardware bring-up.

Artificial Intelligence Intern (Remote) – Code Alpha Aug 2024 – Sep 2024
Remote

- Developed and trained CNN and MLP models using TensorFlow and PyTorch for image classification and related machine learning tasks.

Technical Skills

Programming Languages: C, C++, Python, ARM Assembly, VHDL, Verilog

Data Processing & ML: NumPy, Pandas, SciPy, scikit-learn, TensorFlow, PyTorch

Microcontrollers & Platforms: STM32, PIC18F, AVR, ESP32, Raspberry Pi, Zybo Z7-10, Basys-3 FPGA

Communication Protocols: UART, SPI, I2C, CAN, BLE

Operating Systems: Bare-metal, FreeRTOS, Embedded Linux (basic)

Embedded Security: Cryptography, SCA, CPA, EMFI/FI, DFA.

Development Tools: Xilinx Vivado/Vitis, QuestaSim, STM32CubeIDE, MPLAB X, Altium Designer, MATLAB/Simulink, Git/GitHub

Projects

Phase-Based UHF RFID Localization System

- Developed a Python-based RFID phase measurement algorithm using phase difference to estimate tag positions for indoor localization and access control applications.
- Processing multi-antenna RFID data and evaluating positioning indicators for robust RTLS performance.
- **Tools:** Python, RFID, Wireless Communication, Signal Processing

Flexural Testing Machine (Control PCB + Firmware)

- Designed a 4-layer Altium control PCB and developed STM32 firmware to automate catheter kink testing.
- Implemented stepper and servo motor control, load-cell force acquisition, real-time display via touch HMI, and safety interlocks.
- **Tools:** Altium Designer, STM32, Embedded C, Timers/PWM, I2C, UART

Industrial Radio (Solar-Powered Wireless AC Current Monitor)

- Developed a solar-powered wireless AC current monitoring device using a Rogowski coil interface and ESP32 for industrial environments.
- Transmitted measured current data over LoRa with optional GSM support for remote monitoring.
- **Tools:** Altium Designer, ESP32, LoRa, SPI, Power Management, UART

Industrial Sensor Data Logger (STM32)

- Built an STM32-based data logger that samples sensors via ADC and I2C on a timer schedule, timestamps readings, and streams data over UART or stores it locally.
- Used interrupts for reliable non-blocking acquisition and data handling.
- **Tools:** STM32CubeIDE, Embedded C, ADC, I2C, UART, Timers, Interrupts

Enhanced Cruise Control with Hazardous Road Detection

Final Year Project

- Implemented an adaptive cruise control system on STM32 for a 4WD chassis with hazardous road detection using sensor fusion.
- Integrated ultrasonic and IR sensors, H-bridge motor control, PID-based control, and UART communication.
- **Tools:** STM32CubeIDE, L298N, Ultrasonic Sensors, UART

Additional Information

Languages: English (Professional Working Proficiency), French (Basic Proficiency)

Availability: Open to full-time opportunities starting September 2026; open to relocation within France.